

LLM-Based Assistant for the OpenROAD Project

1. Introduction

OpenROAD is an open-source Electronic Design Automation (EDA) tool that automates the chip design flow from RTL to GDSII. Although it is powerful, it is mostly command-line based and requires users to understand many technical commands and scripts.

New users often find it difficult to:

- Understand OpenROAD commands
- Navigate large documentation
- Debug errors in OpenROAD logs
- Generate correct Tcl scripts for different stages of the design flow

To address these challenges, an **LLM-based AI assistant** can be developed to help users interact with OpenROAD more easily.

2. Proposed Idea

The proposed upgrade is to build an **AI-powered assistant for OpenROAD** that allows users to interact with the system using **natural language**.

The assistant will use the **OpenROAD documentation and GitHub repositories as its knowledge base** and provide intelligent responses to user queries.

The system will function similarly to **ChatGPT but specifically designed for OpenROAD**.

3. Objectives

The main objectives of this system are:

- Simplify the usage of OpenROAD for beginners
- Provide quick explanations of OpenROAD commands
- Generate Tcl scripts for chip design workflows
- Help debug OpenROAD error logs
- Improve accessibility of OpenROAD documentation

4. Key Features

Command Explanation

The assistant can explain OpenROAD commands.

Example:

User Query

"How do I run global placement?"

Response

Explanation of the global_placement command and its usage.

Script Generation

The system can generate **Tcl scripts** required for different stages of the OpenROAD flow such as:

- Floorplanning
- Placement
- Clock Tree Synthesis
- Routing

Error Log Debugging

Users can paste OpenROAD error logs and the assistant will:

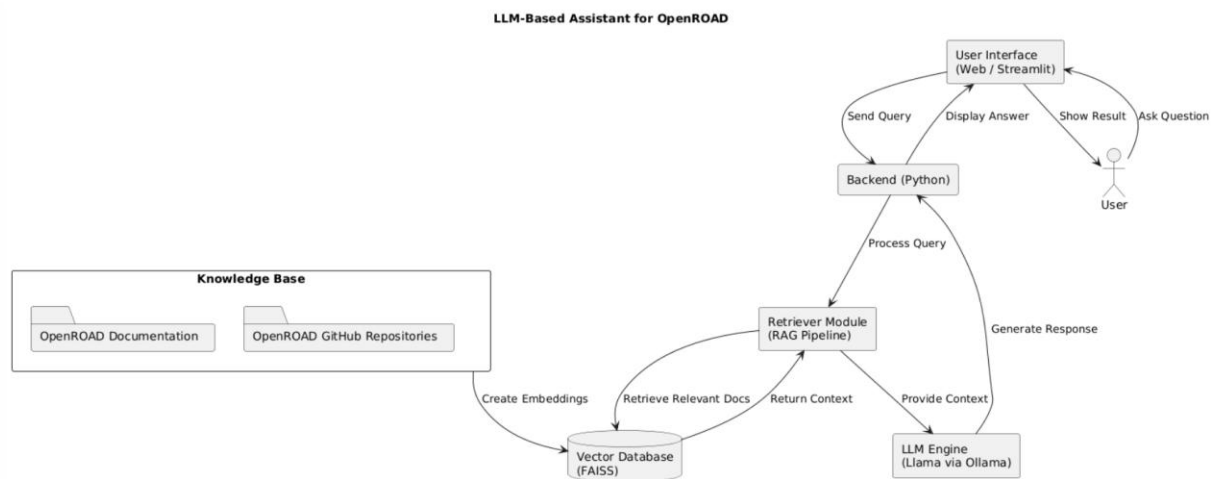
- Explain the error
- Suggest possible solutions
- Provide guidance on fixing the issue

Documentation Search

The assistant will quickly retrieve relevant information from the **OpenROAD documentation** and summarize it in simple terms.

5. System Architecture

The system will follow a **Retrieval-Augmented Generation (RAG)** architecture.



6. Technologies Used

Backend
Python

LLM
Llama model using **Ollama**

Vector Database
FAISS

Embedding Model
Sentence Transformers

7. Expected Outcome

The final system will act as an **AI assistant for OpenROAD**, enabling users to:

- Ask questions in natural language
- Understand OpenROAD commands quickly
- Generate scripts automatically
- Debug errors more efficiently

This will significantly improve the **usability and accessibility of the OpenROAD ecosystem**.